

Varen Sharma

(404)-993-5216 | Atlanta, GA | varensharma10@gmail.com | [linkedin.com/in/varen-sharma](https://www.linkedin.com/in/varen-sharma)

EXPERIENCE

Undergraduate Teaching Assistant - Georgia Institute of Technology, Atlanta, GA **August 2023 – December 2024**
Technology: SQL · NumPy · R · Statistical Data Analysis · ExpertFit · Minitab · SAS

- Assisted in delivery, grading, and execution of Industrial Engineering, Computer Science, and Applied Math courses, and helped over 150 students across 1.5 yrs as a teaching assistant.
- Led and refined grading and feedback procedures for ISYE 4031, Regression and Forecasting. Aided in development of autograder and semester-long project structure used in CS 4400, Database Systems. Designed, delivered, and proctored digital exams and finals in MATH 2552, Differential Equations, for over 300 students.
- Statistically analyzed student results to determine any potential anomalies such as cheating, fraud, or identical copied assignments. Developed training guides for 25+ first-time teaching assistants to use as onboarding materials

Modeling and Simulation Analyst Intern - Georgia Tech Research Institute, Atlanta, GA **May 2023 – August 2024**
Technology: Matlab · NumPy · R · Data Analysis · ExpertFit · Data Conversion · Python

- Led the conversion of radar output data from multiple internal, GTRI modeling, and simulation suites into Government referenced data formats.
- Researched, collected, and curated a commercial dataset using data sourced from client and another 100+ independently verified sources during the project to use for format validation and comparison.
- Spearheaded the ingestion of the modeled radar data into sensor independent products and community tools for signal and image processing.
- Utilized a combination of radar physics, data formats, pre and post processing of data, and industry standards using Matlab and Python to convert radar output data into usable community formats

Procurement and Sales Intern - Urban Outfitters, Philadelphia, PA **May 2022 – August 2022**
Technology: Jupyter · NumPy · R (Programming Language) · Statistical Data Analysis · SQL · Python

- Optimized methods for inventory cataloging, supply and demand management and product placement to better optimize sales patterns and processes.
- Recommended changes to sales model and growth through existing SAP and R analytics software for predicting and refining customer sales patterns.
- Analyzed online and digital order frequency using SQL and CosmoDB data to determine potential seasonal e-commerce trend opportunities across over 100 individual SKUs and implemented more synergistic store displays and promotions.
- Implemented improved front of store and local digital order operations and statistically surveyed their impact using Python, reducing employee idle time by over 23%.

Data Analyst and Instructor – Kumon, Alpharetta, GA **August 2018 – August 2021**
Technology: Excel · NumPy · Linear Regression · Statistical Forecasting · R (Programming Language)

- Analyzed over 75 students' weekly data to determine areas of weakness and strength to tailor individualized progression plans.

- Implemented an overhauled storage system based on historical usage patterns for thousands of worksheets across 20+ levels of difficulty.
- Led focused training of 10+ new employees to modernize teaching and grading methods based on student performance.
- Quantitatively analyzed market demand characteristics to determine center growth opportunities, such as test prep.

SKILLS

Programming Languages: Python, R , Javascript, HTML, Minitab

Databases: SQL, PostgreSQL, Supabase, MSSQL

Tools and Libraries: CPLEX, Gurobi, NumPy, Pandas, PyMySQL, Simio, SAP, Excel, Word, PowerPoint, VBA

Generative AI, Large Language Models: OpenAI, Claude, Deepseek, Operations Research LLM Agents, Prompt Engineering for Operations Research problems

Machine Learning and Data Science: Scikitlearn, Matplotlib, Power BI, Tableau, Lean and Six Sigma Principles

Cloud Platforms: Amazon Web Service, Microsoft Azure

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Bachelor of Science in Industrial and Systems Engineering

Concentration: Operations Research | **Graduation:** May 2025

Honors: Graduated with High Honors, 7× Dean's List, KS2 Technologies Senior Design Award

PROJECTS

Airplane Crash Data Analysis - Georgia Institute of Technology

Jan 2022 – May 2022

Technology: Jupyter · NumPy · Linear Regression · Statistical Data Analysis · Python

- Advanced statistical analysis and hypothesis testing of airplane crash data across a period of 25 years to determine revenue and cost impact of failure.
- Investigated over 100 crashes and their fatality rates for several leading airlines to determine statistical relevance and importance of crashes on flight revenue and operations delays.

Spotify Data Analysis - Georgia Institute of Technology

Jan 2023 – May 2023

Technology: Jupyter · NumPy · Linear Regression · Statistical Data Analysis · Gurobi Optimization · Python

- Analyzed trend of artist popularity versus critical and fan perception across over 250 artists and 700 albums to determine under and over performance relative to perceived music quality.
- Built, analyzed, and verified genre and era specific regression models to capture overarching trends across a sample of over 700 artists' discographies to predict artist and album popularity from perceived quality.

Startup Ideation and Viability Analysis Project - Georgia Institute of Technology

Aug 2023 – Dec 2023

Technology: Jupyter · NumPy · Linear Regression · Statistical Data Analysis · Python

- Designed user interface and minimum capabilities for a food procurement and delivery app focused on connecting users in food deserts to local farmers to address food insecurity.
- Led question design and team execution across 10 rounds of customer surveys and A/B testing to determine ideal landing page and features and gauge product-market fit.
- Analyzed over 150 existing companies to determine sector size and competition to estimate potential company size and valuation through statistical analysis.

Meals On Wheels Senior Design Project- Georgia Institute of Technology

Oct 2024 – May 2025

Technology: PostgreSQL · React · AWS Lambda · Web-Scraping · Python · Cloud App Development

- Led development of an end-to-end cloud user application, hosting inventory data on 70,000+ servings of food, and delivering usage and procurement insights.
- Implemented a spatial and organizational rehaul of inventory storage area, based on item usage velocity, size, and proportional usage, to deliver cost effective systems to manage meal planning, inventory usage, and procurement.
- Final recommendations yielded a potential increase of over 31000 annual meals and a potential 46% reduction in the cost of a single meal.

AWARDS AND INVOLVEMENTS

- IISE Member (Institute of Industrial and Systems Engineers)
- HOPE & Zell Miller Scholarship Recipient
- National AP Scholar
- National Honor Society Member
- National Technical Honor Society Member
- GT Consulting Club
- GT Investment Management Club
- Engineers Without Borders
- Data Science At GT
- Math Modeling Club
- 7x Georgia Tech Dean's List Recipient
- Georgia Certificate of Merit Recipient
- 2025 Georgia Tech ISYE KS2 Technologies Design with Implementation Award